[Assignment-3]

Name : Imtiaz Hossain

ID : 23101137

Section : 05

Course code : CSE 330

Faculty Initial : [AQU]

Ans to the question no 1

$f(x) = x^3 + x^2 - 4x - 4$

a) The exact roots of $f(x) = 0$ are

$x = -2$

$x = -1$ and

$x = 2$

Two fixed-point functions $g(x)$ such that $f(x) = 0$ is equivalent to $x = g(x)$ are:

$$g_1(x) = \frac{x^3 + x^2 - 4}{4}$$

$$g_2(x) = \sqrt[3]{-x^2 + 4x + 4}$$

b) For $g_1(x)$:

$x = -2$,

$\lambda = |g_1'(-2)| = |-2| = 2 > 1$; Diverge

$x = -1$,

$\lambda = |g_1'(-1)| = 0.25 < 1$; Linear convergence

$x = 2$,

$\lambda = |g_1'(2)| = 4 > 1$; Diverge

For $g_2(x)$:

$\lambda = |g_2'(-2)| = 0.67 < 1$; Linear convergence

$\lambda = |g_2'(-1)| = 2 > 1$; Diverge

$\lambda = |g_2'(2)| = 0$; Super Linear convergence

Ans to the question no 2

a) $f(x) = xe^{x} - 1$

$x_0 = 1.5$

we know that,

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

K	x_k	$f(x_k)$
0	1.5	5.7225
1	0.9893	1.6616
2	0.6789	0.3386
3	0.5766	0.0263
4	0.5672	1.567×10^{-4}
5	0.5671	-1.1962×10^{-4}

b) $g(x) = \dfrac{2x+1}{\sqrt{x+1}}$ | $g'(x) = \dfrac{2x+3}{2(x+1)^{3/2}}$

To be super-linearly convergent at a root x_*, we require ~~$g(x_*) = 0$~~ $|g'(x_*)| = 0 = \lambda$

$\therefore \lambda = |g'(x_*)| = 0$

$\Rightarrow \dfrac{2x_* + 3}{2(x+1)^{3/2}} = 0$

$\Rightarrow 2x_* + 3 = 0$

$\therefore x_* = -\dfrac{3}{2}$

Ans to the question no 3

a) $f(x) = x^2 - x + 1$

$x_0 = 0$

we know that,

$$\cancel{f(x} \quad x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

k	x_k	$f(x_k)$
0	0	1
1	1	1
2	0	1
3	1	1
4	0	1

The sequence $0, 1, 0, 1, \ldots$ doesnot converge.

Additionally, $f(x) = 0$ has no real roots, it has imaginary roots: $\frac{1+\sqrt{3}i}{2}$, $\frac{1-\sqrt{3}i}{2}$.

Thus, the sequence fails to approach a root.